

Here's a cleaner, structured, and visually polished version of your notes—easy to revise before exams and quick to scan.

✨ SMT Solving – Clean & Exam-Ready Notes

🧠 1. From SAT → SMT

- **SAT (Boolean Satisfiability)**
Works only with **true/false** variables
- **SMT (Satisfiability Modulo Theories)**
Extends SAT with **real-world domains**:
 - Arithmetic ($\mathbb{Z}$, $\mathbb{R}$)
 - Arrays
 - Strings
 - Functions (EUF)

👉 Goal: Check satisfiability **with respect to a theory**

⚙️ 2. How SMT Solvers Work

SMT = SAT Solver + Theory Solvers

- **SAT Solver**
 - Handles Boolean structure
 - **Theory Solver**
 - Handles domain logic (arrays, arithmetic, etc.)
 - **Architecture: DPLL(T)**
→ Communication between SAT + theory reasoning
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🧩 Key Exercises (Step-by-Step Thinking)

Exercise 1: Sports Interpretation

Formula:

$p(f(f(c, b), g(a, d)))$

✓ Solution Flow:

1. $f(c, b) \rightarrow$ more players $\rightarrow$ **rugby (15)**
2. $g(a, d) \rightarrow$ more popular $\rightarrow$ **football**
3. $f(\text{rugby}, \text{football}) \rightarrow$ more players $\rightarrow$ **rugby**
4. $p(\text{rugby}) \rightarrow$ played with ball? $\rightarrow$ **TRUE**

✓ Final Answer: SAT (TRUE)

Exercise 2: EUF (Equality with Uninterpreted Functions)

Formula:

$p(x, g(f(y), f(z))) \wedge \neg p(y, g(f(x), f(z))) \wedge x = y$

✓ Key Idea:

- Substitute $x = y$

👉 Becomes:

$P \wedge \neg P$

✗ Contradiction

✓ Final Answer: UNSAT

Exercise 3: Array Theory

Formula (simplified view):

- $\text{write}(a, i, x) \neq b$
- $a = b, i = j$
- Reads give same value y

🔑 Axioms:

1. $\text{read}(\text{write}(a, i, v), i) = v$
2. $i \neq j \rightarrow \text{read}(\text{write}(a, i, v), j) = \text{read}(a, j)$
3. **Extensionality**

✓ Reasoning:

- From $i = j$:
 $\text{read}(\text{write}(b, i, x), j) = \text{read}(\text{write}(b, i, x), i) = x$
- Given: $\text{result} = y \rightarrow x = y$

BUT:

- $\text{write}(b, i, x) \neq b \Rightarrow \text{requires } x \neq \text{read}(b, i) \Rightarrow x \neq y$

✗ Contradiction

✓ Final Answer: UNSAT

abc Exercise 4: String Theory

(A) Formula:

$$XY = YX \wedge X \neq Y$$

✓ Possible when strings **repeat a pattern**

Examples:

- $X = "a", Y = "aaaa"$
- $X = \epsilon, Y = "a"$

✓ Final Answer: SAT

(B) Formula:

$$XY = YZ \wedge (\text{no overlap between } X \text{ and } Z)$$

✓ Key Insight:

- If **no overlap**, equation cannot hold

✗ Impossible structure

✓ **Final Answer: UNSAT**

High-Yield MCQ Concepts

Theory Properties

Theory	Key Idea
EU_F	Equality + congruence
LRA	Poly-time
LIA	NP-complete
Arrays	read/write axioms
Bit-vector s	fixed-size + wraparound
Strings	concatenation, word equations

Decidability

✓ **Decidable:**

- EU_F
- LRA, LIA
- Quantifier-free strings

✗ **Undecidable:**

- Non-linear integer arithmetic
- Full arrays
- Full First-Order Logic

SAT vs SMT vs CP

Feature	SAT	SMT	CP
Domain	Boolean	Multi-theory	Finite
Expressiveness	Low	High	Medium
Infinite Domains	✗	✓	✗
Optimization	✗	Limited	✓

Problem-Solving Strategy (Golden Steps)

When you see ANY SMT problem:

♦ Step-by-step:

1. **Identify theory**
 - EUF? Arrays? Strings? Arithmetic?
 2. **Recall axioms**
 - Arrays $\rightarrow$ read/write
 - EUF $\rightarrow$ congruence
 3. **Apply substitutions**
 - Replace using equalities
 4. **Check contradictions**
 - Look for $P \wedge \neg P$
 5. **Use congruence**
 - $x = y \Rightarrow f(x) = f(y)$
 6. **Arrays**
 - Always apply read/write carefully
 7. **Strings**
 - Think **overlap & repetition patterns**
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1-Line Exam Trick

👉 *Most SMT UNSAT proofs = reduce to contradiction after substitution.*

If you want, I can turn this into a **1-page cheat sheet PDF** or give you **MCQs exactly like your exam** (with traps explained).